EV Technology Course — Module 3: Power Electronics (Inverter, DC-DC, Onboard Charger)

Oct 14, 2026 · @Ashishpreet

Three power-electronic boxes convert electricity inside every EV: the traction inverter (battery DC to motor AC), the DC-DC converter (high voltage to 12 V) and the onboard charger (grid AC to battery DC). All three work by switching semiconductors on and off thousands of times a second, and the shift from silicon IGBTs to silicon carbide (SiC) is one of the biggest efficiency gains of the last decade.

1. Learning objectives

- Explain how six switches turn DC into three-phase AC using PWM.
- Compare IGBT, SiC MOSFET and GaN devices.
- Identify conduction and switching losses and how to estimate them.
- Describe the parts of an inverter: power module, gate drivers, DC-link capacitor, sensors, control board, cooling.
- Explain how the DC-DC converter and onboard charger work.
- Describe how power modules and inverters are built and tested.

2. Fundamentals

A power-electronic switch is a semiconductor that is either fully on (low resistance) or fully off (blocking voltage). Operating only in those two states wastes little energy. By switching very fast and varying how long the switch stays on, the converter can shape any voltage it needs. This is **pulse-width modulation (PWM)**: if a switch connected to 400 V is on 60% of the time, the average output is 240 V.

In an inverter, each motor phase connects to a **half-bridge**: one switch to the battery's positive rail and one to the negative rail. Three half-bridges (six switches) produce three sine-shaped phase voltages, 120° apart, at whatever frequency and amplitude the motor controller asks for.

3. Detailed technical explanation

Switch technologies

Device	Material	Typical voltage class	Switching frequency	Strengths	Weaknesses	Typical use
IGBT	Silicon	650–1,200 V	~5–15 kHz	Cheap, mature, robust	Higher switching and part-load losses	400 V inverters, cost-sensitive EVs
SiC MOSFET	Silicon carbide	750–1,700 V	10–50+ kHz	Low losses especially at part load, high temperature, ideal for 800 V	Higher cost; fast switching stresses insulation and EMC	800 V cars, premium 400 V, OBCs
GaN HEMT	Gallium nitride	Up to ~650 V (mostly)	100 kHz–MHz	Very fast; tiny magnetics	Lower voltage, less proven at traction power	OBCs, DC-DC; traction research

Losses

- **Conduction loss:** while on, the switch has a small resistance (MOSFET) or voltage drop (IGBT).
- **Switching loss:** during each on/off transition, voltage and current overlap briefly; this energy is lost every switching event, so it grows with switching frequency.
- SiC switches cut both, especially at light load, which is where cars spend most of their time. This is why SiC can add several percent of range for the same battery.

Inverter anatomy

Part	Job
Power module	Six (or more paralleled) switches on a ceramic substrate, bonded to a cooled baseplate
Gate drivers	Isolated circuits that switch each device on and off cleanly and protect against short circuits
DC-link capacitor (film)	Smooths the battery voltage seen by the switches; absorbs ripple current
Current sensors	Measure phase currents for motor control and protection
Control board	Microcontroller running field-oriented control (Module 5) and diagnostics
Cooling	Water-glycol cold plate or pin-fin baseplate; double-sided cooling in advanced modules

Part	Job
EMC filter, busbars, housing	Contain electrical noise; carry hundreds of amps; protect the electronics

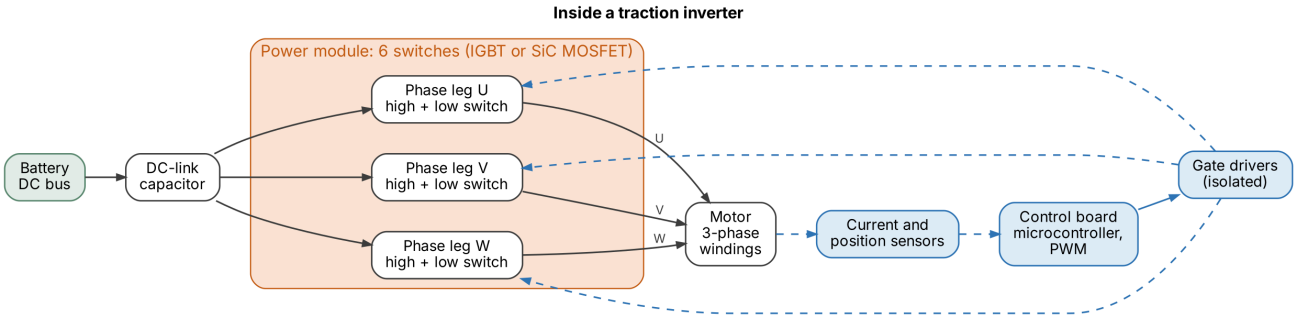
Typical traction inverter efficiency: 97–99%.

DC-DC converter. Steps HV (300–800 V) down to ~14 V to run the 12 V network and charge the 12 V battery; 1.5–3.5 kW in cars. It is **isolated** (uses a transformer) so the LV system has no electrical connection to HV.

Onboard charger (OBC). Converts grid AC to battery DC in two stages: a **power-factor-correction (PFC)** stage that draws clean sinusoidal current from the grid and creates an internal DC bus, then an **isolated DC-DC** stage that adjusts voltage and current to what the BMS requests. Ratings: 3.3, 7.4, 11 or 22 kW. Bidirectional OBCs also support vehicle-to-load (V2L).

Integration trend. OBC + DC-DC + junction box are often combined into one "power box"; inverter + motor + gear into one e-axle (Module 4).

4. Diagram: inside a traction inverter



5. Step-by-step: what the inverter does every PWM cycle

1. The motor controller decides the voltage vector needed (magnitude and angle) from the torque command and rotor position.
2. It converts this into duty cycles for the three phase legs (space-vector PWM).
3. The gate drivers switch the high and low devices in each leg alternately, with a short dead time so both never conduct together (which would short the battery).
4. The motor windings' inductance smooths the pulses into near-sinusoidal currents.
5. Current sensors measure the result; the controller corrects in the next cycle (typically every 50–100 µs).

6. Protection circuits watch for overcurrent, desaturation, overvoltage and overtemperature and shut the switches off within microseconds if needed.

6. Engineering equations

Average voltage from PWM:

$$V_{avg} = D \times V_{dc}$$

Maximum line-to-line RMS voltage with space-vector PWM (linear range):

$$V_{LL,rms,max} \approx \frac{V_{dc}}{\sqrt{2}}$$

Conduction loss (MOSFET) and switching loss:

$$P_{cond} = I_{rms}^2 R_{ds(on)} \qquad P_{sw} = (E_{on} + E_{off}) \times f_{sw}$$

Junction temperature:

$$T_j = T_{coolant} + P_{loss} \times R_{th(j-c)}$$

Converter loss from efficiency:

$$P_{loss} = P_{out} \left(\frac{1}{\eta} - 1 \right)$$

D = duty cycle (0–1), V_dc = battery voltage, R_ds(on) = on-resistance (Ω), E_on/E_off = energy per switching event (J), f_sw = switching frequency (Hz), R_th = thermal resistance (K/W).

7. Worked examples

Example 1: available motor voltage. On a 400 V bus: V_LL,rms max ≈ 400 ÷ √2 = **283 V**. On 800 V, double. This limits base speed (Module 2).

Example 2: conduction loss. Six SiC switches, R_ds(on) = 4 mΩ each, 150 A rms through each: 6 × 150² × 0.004 = **540 W**.

Example 3: switching loss, SiC vs IGBT at 10 kHz. SiC: 4 mJ per cycle → 0.004 × 10,000 × 6 = **240 W**. IGBT: 12 mJ per cycle → **720 W**. (Illustrative values; real values depend on current, voltage and device.)

Example 4: junction temperature. 150 W in one switch, R_th = 0.25 K/W, coolant 65 °C: T_j = 65 + 37.5 = **102.5 °C**, well below a 175 °C limit.

Example 5: DC-DC current. 2 kW delivered at 12 V: I = **167 A** on the LV side. This is why LV cables are thick.

8. Real-world EV examples

- **Tesla Model 3 (2017):** first mass-market car with a SiC traction inverter.
- **Hyundai E-GMP (Ioniq 5, EV6):** 800 V with SiC in the rear inverter.
- **BYD:** in-house SiC power modules; its 1,000 V Super e-Platform (March 2025) uses SiC devices rated up to 1,500 V.
- **Indian market:** car inverters mostly from global tier-1s or OEM in-house designs; many two-wheeler controllers are designed and assembled in India, while power semiconductors are imported.

9. Manufacturing applications

- **Power module assembly:** die attach (solder or silver sintering), wire bonding or copper clips, substrate (DBC/AMB ceramic) to baseplate, gel or epoxy encapsulation. Sintering and clips improve life under thermal cycling.
- **Control boards:** SMT lines, automated optical and X-ray inspection, conformal coating.
- **Final assembly:** busbar welding, capacitor fitting, cold plate sealing (leak test), potting where needed.
- **End-of-line tests:** hi-pot and insulation, functional run on a load bench, efficiency spot checks, EMC sampling.
- **Costing:** semiconductors are the largest cost line (SiC dies are expensive), then capacitor, gate drivers, cooling, housing. Watch die price trends: SiC prices have fallen as 200 mm wafers ramp.

10. Common mistakes and misconceptions

- **"An inverter just flips DC to AC."** It synthesises variable-frequency, variable-voltage AC under closed-loop control.
- **"Higher switching frequency is always better."** It reduces motor current ripple and noise but raises switching loss and EMC challenges.
- **"SiC only matters for 800 V."** It also improves 400 V efficiency at part load.
- **"The DC-DC is trivial."** If it fails, the 12 V battery drains and the car shuts down.

11. Practical case study: inverter failures from thermal cycling

Problem. A fleet of electric buses on hilly routes sees inverter failures after 3 years, well before design life.

Root cause. Repeated hill climbs cause large junction-temperature swings; aluminium bond wires on IGBT dies lift off (fatigue), raising resistance until devices fail.

Corrective action. Derate peak current when temperature swings are large, improve coolant flow, change to a module with sintered die attach and copper clips, and use mission-profile-based lifetime testing (power cycling) in validation.

Lesson. Inverter life is set by thermal cycling, which depends on the duty cycle as much as on the design.

12. Revision notes

- Six switches + PWM make three-phase AC; dead time prevents shoot-through.
- Losses = conduction + switching; SiC cuts both, especially at part load.
- Inverter: power module, gate drivers, DC-link capacitor, sensors, control board, cooling.
- DC-DC: isolated HV → 12 V; OBC: PFC + isolated DC-DC.
- Life limited by thermal cycling of the power module.

13. Glossary

Term	Meaning
PWM / SVPWM	(Space-vector) pulse-width modulation
IGBT	Insulated-gate bipolar transistor
SiC MOSFET	Silicon carbide metal-oxide-semiconductor field-effect transistor
GaN	Gallium nitride
Half-bridge	High and low switch pair for one phase
Dead time	Short pause so both switches are never on together
DC link	Capacitor bank at the inverter input
PFC	Power-factor correction

14. Knowledge check

Conceptual

1. Why do power switches run either fully on or fully off?
2. What happens if both switches in a half-bridge turn on together?
3. Why does SiC improve range most at light load?
4. Why must the DC-DC converter be isolated?
5. What wears out power modules?

Numerical

1. An 800 V bus switched at 60% duty: average voltage?
2. An inverter delivers 100 kW at 98% efficiency. How much heat must cooling remove?
3. A DC-DC supplies 2.5 kW at 14 V from a 400 V bus at 95% efficiency. Output current and input current?

Application

1. Build a cost comparison framework for IGBT vs SiC inverters at vehicle level.
2. An OEM wants V2L. What changes in the OBC?

15. Answer key

Conceptual

1. In those states either voltage or current is near zero, so power loss (V × I) is small; in between, both are large.
2. A direct short across the battery (shoot-through): huge current, device destruction.
3. IGBTs have a fixed voltage drop and high switching energy that dominate at low current; SiC's resistive behaviour and low switching energy keep losses small there.
4. For safety: LV parts that people touch must have no conductive path to HV.
5. Temperature cycling: different expansion rates of silicon, solder and copper fatigue bond wires and die attach.

Numerical

1. 0.6 × 800 = **480 V**.
2. P_loss = 100 × (1/0.98 − 1) = **2.04 kW**.
3. Output: 2,500 ÷ 14 = **179 A**. Input: 2,500 ÷ 0.95 ÷ 400 = **6.6 A**.

Application

1. Inverter cost delta (SiC dies vs IGBT) against savings: smaller battery for the same range (kWh saved × $/kWh), smaller cooling, possibly lower cable mass; plus range gain value. Include warranty and supply risk.
2. The OBC must be bidirectional (active rectifier/inverter stage instead of a diode bridge, bidirectional DC-DC), with software for grid-quality AC output, plus an AC outlet or adapter and safety interlocks.

16. Further learning

- Mohan, Undeland & Robbins, *Power Electronics: Converters, Applications, and Design*.
- Infineon, Wolfspeed and STMicroelectronics application notes on traction inverters (free).
- NPTEL, "Power Electronics" (IIT).
- US DOE / ORNL inverter benchmarking reports.

Next: Module 4, e-drive integration.